question how science works. We've seen how it doesn't work—it's not about verification or purely empirical statements, as the positivists thought; it's not about falsifiability, as Popper thought; nor is it about being completely independent from the complex historical forces that have buffeted and buoyed the characters we've met over the course of this book. So how does science work? Echoing the end of Chapter 11, that's a fabulously complex question. The long answer would take another book. But the short answer is that science involves a combination of experiment, mathematical and logical reasoning, unifying explanations, and biases that scientists bring to the table from their own lives and the cultures they live in. We work to reduce those biases; we don't always succeed, but the explicit attempt to account for and reduce those biases is an important part of the process, properly conducted. The whole edifice of science is geared toward this goal. And, given the phenomenal explanatory power and predictive success of science, it would be foolish in the extreme to give scientific truths no more credence than idle speculation, religious articles of faith, or deeply held cultural values. Science, done right, works hard to respect absolutely no authority at all other than experience and empirical data. It never succeeds entirely, but it comes closer and has a better track record than any other method we apes have found for learning about the world around us, a world we never made.



The story of the search to understand quantum physics is an emphatically scientific one. Yet the cultural and historical forces at work over the course of this book, while par for the course, are still troubling. How can we distinguish a controversy like the one over quantum foundations—a legitimate scientific controversy if ever there was one—from a manufactured pseudo-controversy about science, like the "debates" over evolution, global warming, and homeopathy? It might seem tempting to compare them, after all: from the perspective of someone who believes (erroneously) that climate change isn't real, evolution didn't happen, and homeopathy works, these are all stories of an overwhelming scientific consensus fought by a scrappy minority of independent thinkers dedicated